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DETAIL SPECIFICATION

THIMBLE, SYNTHETIC ROPE

This specification Is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers thimbles used with synthetic rope.

1.2 Classification. Thimbles shall be the sizes used with 1-1/8 to 10 inch circumference synthetic rope shown on figure 1, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection By Attributes (DoD adopted)

(Copies of this document are available for purchase online at https://www.asq.org.)

ASTM INTERNATIONAL

ASTM B148	-	Standard Specification for Aluminum-Bronze Sand Castings.

(Copies of this document are available for purchase online at https://www.astm.org.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, Attn: VAI, 3990 East Broad Street, Columbus, OH 43218-3990 or fluidflow@.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online at https://assist.dla.mil.

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3. REQUIREMENTS

3.1 Design. The thimble shall be a one-piece aluminum bronze casting conforming to ASTM B 148, grade D, as-cast.

3.2 Thimble strength. Thimbles shall be of sufficient strength to pass the tests specified in 4.1.3.

3.3 Dimensions. Thimbles shall conform to the dimensions shown on figure 1, as specified (see 6.2).

3.4 Identification. Each thimble shall be marked for identification by giving the rope size circumference and the wording “For Synthetic Rope” as shown on figure 1 (see 6.2 and 6.3). Figures shall be depressed in the casting. Rope circumference may be shown on one side, and the lettering on the other side, when the surface area does not permit all figures on one side.

3.5 Workmanship. The surface of the thimble shall be free from fins and burrs, to prevent snagging of the rope. Groove surfaces for the rope shall be a roughness no greater than 125-microinch finish. Edges shall be rounded full radius.

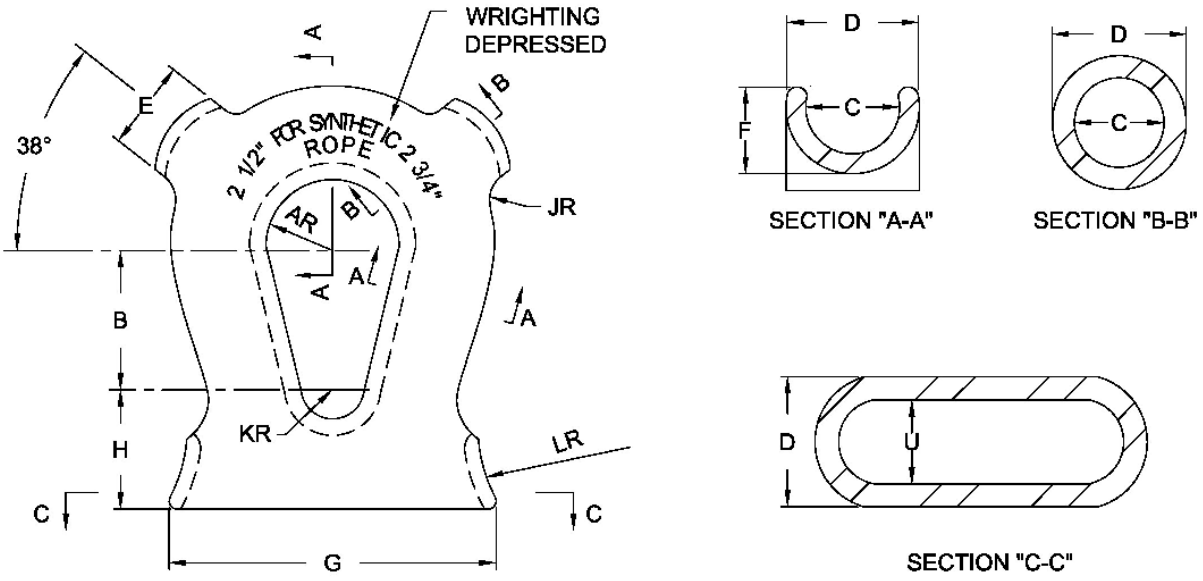


FIGURE 1. Thimble sizes and dimensions

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Rope diam. (reference)	Rope circum.	2/A	2/B	1/C	1/D	2/E	2/F	2/G	2/H	2/J	2/K	2/L	Anchor shackle size	Approx weight pounds
3/8 - 7/16	1-1/8 - 1-1/4	7/16	13/16	9/16	3/4	1/2	1/2	2-1/16	5/8	3/16	3/16	1	3/8	0.3
1/2 - 9/16	1-1/2 - 1-3/4	1/2	1	11/16	29/32	9/16	5/8	2-3/8	7/8	1/4	7/32	1-1/16	7/16	0.5
5/8 - 3/4	2 - 2-1/4	9/16	1-3/16	7/8	1-1/8	5/8	3/4	3-1/8	0.1	9/32	9/32	1-1/8	9/16	0.6
13/16 - 15/16	2-1/2 - 2-3/4	5/8	1-7/16	1-1/16	1-3/8	3/4	13/16	3-3/4	1-3/16	3/8	5/16	1-3/16	5/8	0.9
1	3	13/16	1-9/16	1-3/16	1-9/16	1	7/8	4-1/4	1-3/8	7/16	13/32	1-1/4	3/4	1.6
1-1/8 - 1-1/4	3-1/2 - 3-3/4	15/16	1-7/8	1-7/16	1-27/32	1-1/16	1	5-1/16	1-9/16	1/2	15/32	1-5/16	7/8	1.9
1-5/16	4	1-1/16	2	1-1/2	1-15/16	1-1/8	1	5-1/2	1-3/4	1/2	17/32	1-3/8	1	2.6
1-1/2	4-1/2	1-3/16	2-1/4	1-11/16	2-5/32	1-3/16	1-1/8	6-1/4	1-7/8	1/2	5/8	1-7/16	1	3.2
1-5/8	5	1-5/16	2-7/16	1-13/16	2-5/16	1-1/4	1-1/4	6-3/4	2-1/16	1/2	23/32	1-1/2	1-1/8	4.8
1-3/4	5-1/2	1-7/16	2-5/8	2	2-17/32	1-5/16	1-3/8	7-3/8	2-5/16	3/4	13/16	1-5/8	1-1/4	7.8
2	6	1-5/8	2-7/8	2-1/4	2-13/16	1-3/8	1-1/2	8-3/8	2-1/2	1	29/32	1-11/16	1-1/4	8.5
2-1/8	6-1/2	1-3/4	3-1/16	2-3/8	3	1-7/16	1-5/8	9	2-11/16	1	1	1-3/4	1-3/8	9.5
2-1/4	7	1-15/16	3-1/4	2-1/2	3-1/8	1-1/2	1-3/4	9-3/8	2-7/8	1	1-1/8	2	1-1/2	10.6
2-5/8	8	2-3/16	3-11/16	2-7/8	3-5/8	1-5/8	1-7/8	10-5/8	3-1/4	1	1-1/4	2-1/8	1-5/8	18.5
3	9	2-7/16	4-1/8	3-1/4	4-1/8	1-3/4	2	11-3/4	3-5/8	1-1/4	1-3/8	2-1/4	2	24.0
3-1/4	10	2-11/16	4-9/16	3-1/2	4-1/2	1-7/8	2-1/4	13-1/8	4	1-1/2	1-1/2	2-1/2	2	38.0

1/ Tolerance on dimensions C and D +0 / -3 percent.
2/ Tolerance on all other dimensions ± 8 percent.

FIGURE 1. Thimble sizes and dimensions.- Continued

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4. VERIFICATION

4.1 Conformance inspection.

4.1.1 Sampling.

4.1.1.2 Lot. Thimbles of the same size offered for delivery at the same time shall be considered a lot.

4.1.1.2 Sampling for examination and tests. A random sample of thimbles shall be selected from each lot in accordance with ASQ Z1.4, for examination and tests, as specified in 4.1.1.2.1 and 4.1.1.2.2

4.1.1.2.1 Sampling for examination. A random sample of thimbles shall be selected from each lot with normal inspection for the examination specified in 4.1.2.

4.1.1.2.2 Sampling for strength testing. A sample of thimbles shall be randomly selected with an inspection level S-4 for tests specified in 4.1.3.

4.1.2 Examination. Each of the sample thimbles selected in accordance with 4.1.1.2 shall be visually and dimensionally examined to verify compliance with this specification. Examination shall be conducted as specified in table I. Any thimble in the sample containing one or more defects shall not be offered for delivery. If the number of defective thimbles in any sample equals or exceeds the acceptance number for that sample, as specified in 4.1.1.2.1, it shall be cause for rejection of the lot represented by the sample.

TABLE I. Classification of defects.

Categories	Defects
Critical:	Thimble does not pass load tests.
Major:	
101	Thimble material, not as specified (see 3.1).
102	Dimensions not in conformance with figure 1 within specified tolerances.
103	Groove for rope does not have the specified surface finish (see 3.5).
104	Thimble edges sharp and not rounded as specified (see 3.5).
105	Identification figures cast in thimble are not legible.
106	Casting is porous and has imperfections.
Minor:	None.

4.1.3 Strength tests. Sample thimbles shall be tested by the application of loads specified in table II. There shall be no permanent deformation or cracks as a result of the strength test. If the number of thimbles equals or exceeds the rejection number specified in 4.1.1.2.2, the lot shall be rejected.

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TABLE II. Loading for strength tests.

Synthetic rope circumference	Test load
(Inches)	(Pounds)
1-1/8 – 1-1/4	4,900
1-1/2 -1-3/4	7,700
2 - 2-1/4	12,600
2-1/2 – 2-3/4	17,600
3	24,200
3-1/2 – 3-3/4	36,300
4	41,200
4-1/2	50,600
5	62,700
5-1/2	74,800
6	89,000
6-1/2	99,000
7	121,000
8	150,700
9	187,000
10	222,000

4.1.3.1 Test shall be performed in a testing machine using synthetic rope to simulate actual working conditions.

4.1.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The thimbles covered by this specification are intended for use with synthetic rope, with the exception of emergency towing application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and quantity of thimbles (see 1.2 and 3.3).
(c) Identification markings required (see 3.4).
(d) Packaging requirements (see 5).

6.3 Identification. In marking the thimble (see 3.4), the wording "For Plastic Rope" may be used until the tooling or pattern is required to be replaced.

6.4 Subject term (key word) listing.

Aluminum
Bronze
Casting
Groove

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - AT
DLA - CC

Preparing activity:
DLA - CC

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